



ashwa RACING

NEWSLETTER - July 2017



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Manufacturing phase for the 2017-18 race season

The designs of RZ-X8C & RZ-X8H have been finalised thereby marking the beginning of the manufacturing phase for the 2017-18 race season.

Following the procurement of steel tubes in the month of May, the tubes were laser cut and profiled in accordance to the requirements of the chassis design, at Magod Laser Machining Pvt Ltd.



The vent holes required on the tubes to facilitate the welding process, were drilled in-house, by the team.



Raw materials for the fixtures were procured from Mallar Group, to fabricate the chassis.

The fixtures were milled at Venkateshwara Manufacturing Systems, Venus Engineering Industries and Sun Precision.

The final process - DRO drilling of the fixtures was completed at Accu precision.

Magod Laser Machining have also extended their support to the team by manufacturing the sheet metal fixtures.

The next few days will be dedicated towards the fabrication of the chassis on the base plate.

The team is extremely grateful to all its manufacturing partners for extending their support to Ashwa Racing for the 2017-18 Race season.

The Ashwa Racing Internship Program

As a part of the PR activities, Ashwa Racing has always wanted to disseminate its work culture to a young school-going audience via workshops and internships. The objective behind this is to help youngsters in school, make an informed choice about their career.

This summer, the team was approached by Yash Tiwari, a 17 year old boy entering 12th grade, who was interested in Science and Engineering. To obtain clarity and insight about the daily life of an Engineer, Yash approached Ashwa Racing, seeking a week long internship. This gave the team the perfect opportunity to start The Ashwa Racing Internship Program, where high school students are allowed to intern and attend workshops, to familiarize them with the Engineering environment and workings of Ashwa Racing.

In his time here, Yash learnt about the basics of managing oneself in a professional setting, managing a team and some technical aspects of building a racecar. The team strongly believes that this undertaking has helped in giving something back to the community, in a way that is unique to Ashwa Racing. It was wonderful having a young enthusiastic learner trying to gain as much knowledge as possible, in a limited time, while also helping some of the team members become better teachers and better leaders. The team intends to do more of the same in the coming years.

Engine Upgrade for the 2017-18 Hybrid Race Season



For the 2017-18 race season, the team has decided to use a CBR 250R engine on the hybrid prototype - RZ-X8H. The previous hybrid car used a Pulsar 220 DTS-i engine. This change has been adopted, as the CBR 250R produces more power which implies that even a larger battery pack can be used and charged efficiently. Further, the liquid cooled system in the engine is far superior to the oil cooled system on the Pulsar 220 DTS-i.

The engine was run successfully on the test rig, giving the team the confidence to proceed with further tuning of the engine.



RZ-X7C

The race-car designed by the team for the 2016-17 race season under the combustion category is the RZ-X7C which is powered by the Suzuki GSXR 600cc engine with a power output of 75 bhp.

The prototype which weighs 212 kg, having a suspension system with unequal A-arms and ZF dampers, is capable of accelerating from 0 to 100 kmph in 4.8 seconds.

The prototype features pneumatic

shifters with a hydraulic clutch and rpm lights for precise shifting.

Onboard display showing gear position, rpm, engine temperature and battery voltage has also been added to the prototype to enhance control and dynamic performance of the car.

The main design goal in building this race-car is the performance and reliability whilst attaining optimum power to weight ratio.

Testing makes a car perfect

Sleepless nights were spent ensuring continuous and comprehensive testing of the car was done for 3 months, prior to the event which has helped the team understand the strengths and the shortcomings of the car. Rework, adjustments and modifications were made to overcome the failures to ensure that the prototype is all set to excel at the event.

Bon Voyage



After a rigorous period of testing, analysis and rework, the event-ready combustion prototype RZ-X7C was shipped on the 20th of June, 2017 along with the required spare parts, necessary tools and components in the cargo box crafted by Bhumi Putra Architecture.

The cargo is scheduled to reach Italy on the 9th of July 2017, from where the car would be taken to the tracks of Verano De Melagari for the most awaited event - Formula Student Italy, 2017.

Ashwa Racing would like to thank IMPEL CARGO LINES for supporting the team's endeavour and coming on board as its Sailing Partner, thereby helping the team overcome a major hurdle - Logistics, with considerable ease.

Care beyond Cargo



Impel Cargo is a world-class provider of ocean and air consolidations solutions, based off of Chennai, India. They have an extensive network of clients from over 100 countries and a team possessing over 40 years of experience in the field of logistics.

Impel Cargo goes beyond the conventional part of logistics, reminding the customers to go green, when they can, keeping in mind the environment, health, social responsibility, security and ethics.

The team is extremely delighted with the service provided by Impel Cargo Lines and is grateful to Karthik Sir, Venkatesh Sir and the entire team of Impel Cargo who have taken up this consignment with so much attention.

Ashwa Racing is honored to be associated with Impel Cargo who truly prioritise the relationship they build with the client, over the financials involved.

The team hopes to continue this relationship with Impel Cargo, in the future years to come.



SPONSOR FOCUS

SUN PRECISION TOOLS

Sun Precision has supported us with milling of the fixture blocks to be used on the combustion chassis thereby allowing us to dedicate funds towards other technical developments. We profoundly thank them for their support in helping us realise our chassis design.

Middleby India is an in-kind sponsor who has supported us with water jet cutting for a large number of our aluminium components. We highly value the timely delivery of the components. We thank them for their support and hope to continue this relationship in the future years to come.



Srinu Engineering Industries

We are extremely grateful to Srinu Engineering Industries, who have helped us with CNC milling of our steering and suspension components, namely: the rack casings, arb housings, casing mounts and C- Clamps.

Meckon is a manufacturer of precision machined components, jigs and fixtures. We thank them for their support in manufacturing drive train adapters for the 2017-18 race season.

MECKON



Venus has supported us with milling of the fixture blocks of the hybrid and combustion chassis. This is part of the fixture setup to initiate fabrication of the chassis. We thank Venus for their much needed and timely support.

SPONSOR FOCUS

SHRI VENKATESHWARA HI-TECH

S. V. Hitech has been an in-kind sponsor to Ashwa Racing, since 2015, helping us manufacture components using their advanced CNC Machines ensuring highest quality standards. Their continued support in manufacturing the differential mounts for the 2017-18 race season has ensured that our quest to make our designs lighter is not constrained by manufacturing challenges.

Venkateshwara Manufacturing Systems has supported us with milling of the fixture blocks of the hybrid chassis which are used to support critical tubes of the chassis on the base plate. We thank them for their support and encouragement towards the project.

VENKATESHWARA MANUFACTURING SYSTEMS



Accu-Precision is a manufacturer of custom parts, sheet metal and assemblies for customers in aerospace and industrial sectors. We thank them for helping us out with DRO drilling of the fixtures required to fabricate the chassis.

Hi-Flex Technologies is a manufacturer of precision machined components and precision turned components. We thank them for supporting us with manufacturing of few suspension components including pillow ball housing and brass inserts for the welding of A-arms.

HI-FLEX TECHNOLOGIES



We thank Mallar Group for their continued support in helping us with the milling of our baseplate, on which our chassis is built. We also thank them for helping us with procurement of material for the fixtures.

AUDIT - June 2017



R. V. COLLEGE OF ENGINEERING

अश्वरACING

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R. V. COLLEGE OF ENGINEERING **अश्वर** RACING

Accounts for the period from 1-06-2017 to 30-06-2017

RECIEPTS	AMOUNT
Opening Balance in Kotak Mahindra Bank RVCE Branch	
SB A/c 136010046665	₹ 1,924,568.68
SB A/c 136010058514	₹ 677,242.12
SB A/c 136010060793	₹ 520,966.93
cash in hand	₹ 1,460.00
45H, 59C remaining cash in hand	₹ 182.00
MR aluminium refund	₹ 4,505.00
sponsorship money	₹ 110,000.00
italy flight ticket	₹ 259,000.00
RZX7C batch visa money	₹ 5,362.00
RZX7C batch seed money	₹ 9,004.00
RZX6C batch seed money	₹ 4,000.00
TOTAL	₹ 3,516,290.73

PAYMENTS BY	AMOUNT
Powertrain & Smp	₹ 12,966.00
Electrical	₹ 2,197.00
Miscellaneous	₹ 1,004,512.50
Brakes & Logistics	₹ 559,121.00
Chassis & Suspension	₹ 54,486.00
SMS & Wire Charges	₹ 723.09
Cash In Hand	₹ 1,190.00
Closing balance in Kotak Mahindra Bank , RVCE Branch	
SB A/c 136010046665	₹ 1,304,250.41
SB A/c 136010058514	₹ 344,877.80
SB A/c 136010060793	₹ 231,966.93
TOTAL	₹ 3,516,290.73

Utkarsh

Utkarsh Agrawal
Finance Team

अश्वर RACING

R.V. College of engineering

ALUMNUS OF THE MONTH



R Manjeeth, AKA Manny has been a part of the Ashwa Racing team since 2011. During his tenure, he was a member of the engine subsystem as well as the Sponsorship, marketing and PR module.

Manny, a man of few words, was well known for his calm demeanour even during tense situations. When the going gets tough, Manny gets going and then the going gets easy. His creative approach to solving problems, both theoretical and practical, proved to be vital during critical phases of the project.

Even when the team was confronted with difficult and messy problems like fuel tank leaks and situations alike, he took on the challenges with a smiling face.

He has a vast knowledge base in many areas of engineering such as vehicle powertrains, theoretical

computational fluid dynamics, the thermal sciences and manufacturing processes, to name a few. His work on the design and fabrication of intake manifolds is something which every current batch refers to for guidelines because of the simplicity of the manner in which he addressed a very complicated engineering problem.

He is greatly admired by all junior members who have had the opportunity to work under his guidance.

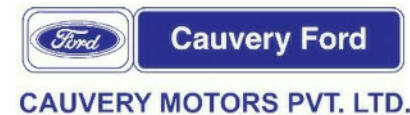
As a member of the sponsorship, marketing and PR team, he exhibited a lot of discipline and good conduct while interacting with potential and current sponsors. Manny has been instrumental in securing several financial and in kind sponsors. Although based in Bangalore, he even made the effort to travel all the way to Mumbai, Pune, Chennai and even Delhi to make partnerships materialize, on many occasions, sometimes even on short notice. Such commitment was rare. In addition to all these responsibilities, he even looked into matters related to procurement of important OEM components.

Presently, Manny is a member of the Alumni council of Ashwa Racing where he is a mentor to the members of the engine subsystem. The team is extremely thankful to him for all his support.

2016-17 COMBUSTION PARTNERS



2016-17 SERVICE PARTNERS



2016-17 COMBUSTION - PRODUCT PARTNERS

**BHUMIPUTRA
ARCHITECTURE**

Continental 

 **MOUSER
ELECTRONICS**

  **FAG**
SCHAEFFLER GRUPPE


MY OBJECTIFY

molex[®]

 **STAHLWILLE**[®]

MOTUL

 bhatia aerospace
 bhatia tools & technology

UNIFLEX

 **TE**
connectivity

FESTO

LOCTITE[®]

BAKER
....a passion for precision



**MANN +
HUMMEL**

 **Suprajit**

 **MIRANDA TOOLS**

2016-17 COMBUSTION - MANUFACTURING PARTNERS



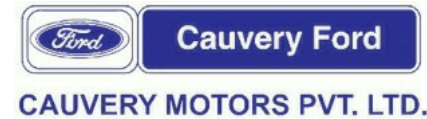
DMG MORI



MALLAR
GROUP



www.icp-india.com



EIGEN ENGINEERING SERVICES

PRA/GATI

LIEBHERR



Srinu Engineering
Industries

SANSEARA
ENGINEERING



HI-FLEX TECHNOLOGIES

AGNI MOTORS



MECKON



auma
Solutions for a world in motion

VENKATESHWARA MANUFACTURING SYSTEMS

SUN PRECISION TOOLS

2016-17 HYBRID PARTNERS

Life Is On

Schneider
Electric

AMS[®]
In Pursuit of Excellence

BFW

infineon

PEGASYS

سابك
sebia

DURAG

**GRIND
MASTER**

DANA

SPICER[®]
Drivetrain Products

SOLVAY
asking more from chemistry[®]

Phillips
MACHINE TOOLS INDIA

Werner Finley[®]
Process Cooling Since 1986

DuFlon[™]
Engineering Excellence

BILL FORGE
YOUR PROACTIVE PARTNER

RÖHM
driven by technology

2016-17 SERVICE PARTNERS

Wrike

RICARDO



ANSYS[®]

joe's
CUSTOMERS | PERFORMANCE TUNING | SERVICE

SRV
SOLUTIONS INC

MERITOR

RACE dynamics
Performance Through Innovation

**MANN +
HUMMEL**

2016-17 HYBRID- PRODUCT PARTNERS



BUILDING TRUST



molex®

LOCTITE®

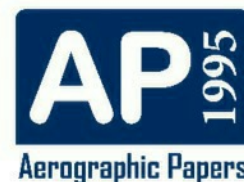
MOTUL

element14

CONTROLWELL



UNIFLEX



2016-17 HYBRID - MANUFACTURING PARTNERS



DMG MORI



MALLAR
GROUP



Fiè TECH
Fantasy in Fibreglass



Cauvery Ford

CAUVERY MOTORS PVT. LTD.



EIGEN ENGINEERING SERVICES

PRAAGATI

LIEBHERR



BHAGAVATHI
TOOLINGS PVT. LTD.



SHRI VENKATESHWARA HI-TECH

Srinu Engineering
Industries



MANUCHO MACHINES (P) LIMITED
ISO 9001-2008 COMPANY

HI-FLEX TECHNOLOGIES



AGNI MOTORS



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Solutions for a world in motion



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Chassis : Brian N Quadras	Workshop : Arunraj Subbaraj
Suspension : Kanal Sebastian	Admin : Arunraj Subbaraj
Brakes : Vamsi Naramsetty	Logistics : Pallav Rawat
Steering : Eshwarmurthy T	Finance : Mohammad Hannan
Engine : R Manjeet	SMP : Karan Chatrath
Drivetrain : Rajat Kabra	HR : Tejas Babaria
Electrical : Siva Subramanian	Testing : Rajit Malik
Coding : Arunraj Subbaraj	IT : Arunraj Subbaraj

STATIC EVENT MODULE COORDINATOR

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